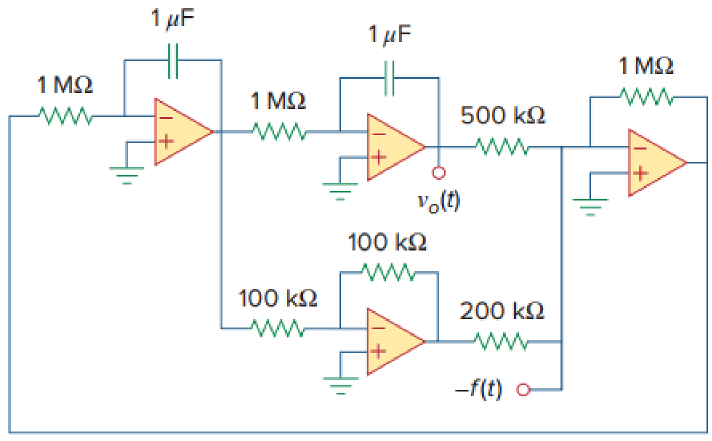
Chapter 6, Problem 80P

Problem

Figure 6.93 presents an analog computer designed to solve a differential equation. Assuming *f*(*t*) is known, set up the equation for *f*(*t*).

Figure 6.93:



Step-by-step solution

Step 1 of 4

First we see that there are 4 op-amps, in which are two are integrating op-amps and two are inverting op-amps.

For the integrating op-amps if the input is v_i then the output would be $v_o = -\frac{1}{RC}\int v_i\, dt$

For the inverting op-amps if the input is v_i then the output would be $v_o = -\frac{R_F}{R_i} v_i$

Here we are given that the integrating op-amp in the center has an output of $v_o(t)$.

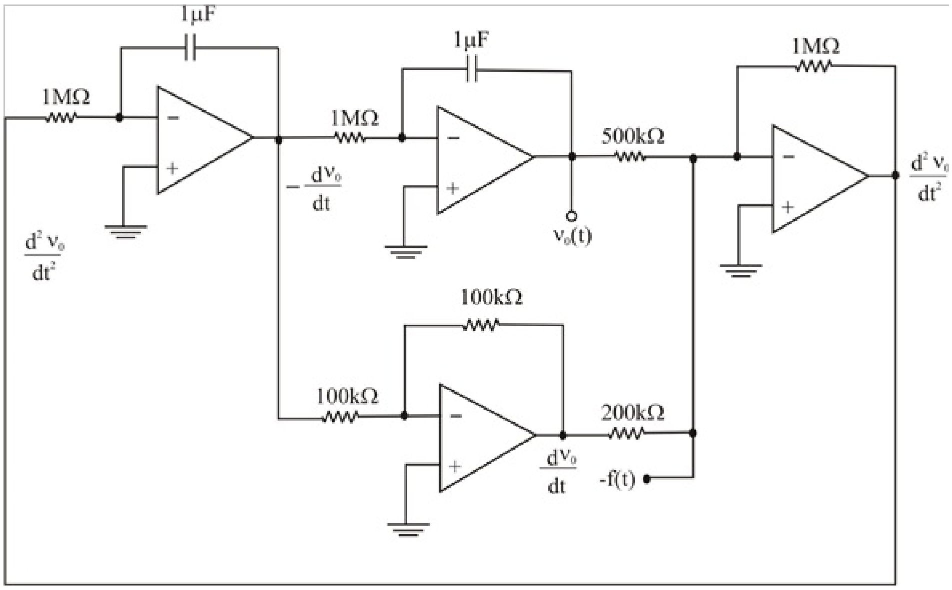
Hence its input should be $-\frac{dv_o}{dt}$

Therefore input to the integrating op-amp should be $\frac{d^2 v_o}{dt^2}$.

Since the input of the integrating op-amp is the output of the inverting op-amp.

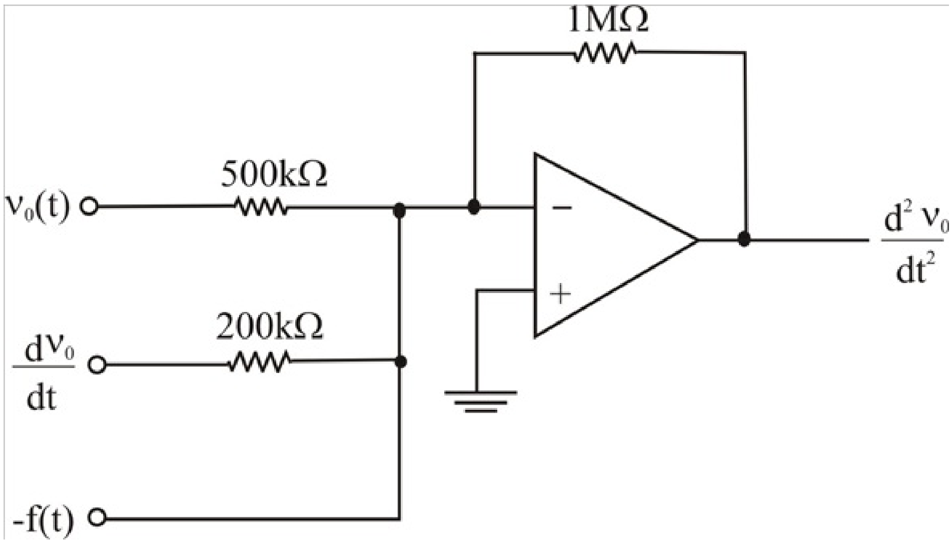
Step 2 of 4

We shall carefully redraw the circuit as,



Step 3 of 4

Hence if we draw the only inverting op-amp



Step 4 of 4

This is a summing amplifier, therefore the output equation is

$$\frac{d^2v_o(t)}{dt^2} = -\left[\frac{1\times10^6}{500\times10^3}v_o(t) + \frac{1\times10^6}{200\times10^3}\frac{dv_o}{dt} - f(t)\right]$$

$$\frac{d^2v_o(t)}{dt^2} + \frac{1\times10^6}{200\times10^3}\frac{dv_o(t)}{dt} + \frac{1\times10^6}{500\times10^3}v_o(t) = f(t)$$

$$\frac{d^2v_o(t)}{dt^2} + 5\frac{dv_o(t)}{dt} + 2v_o(t) = f(t)$$

$$\therefore \boxed{\frac{d^2v_o}{dt^2} + 5\frac{dv_o}{dt} + 2v_o = f(t)}$$